

Clifford algebras over the field-cores of combinatorial games, and the Arf invariant as a win-bias

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Abstract

Conway’s combinatorial games form a partially ordered abelian group under disjunctive sum but not a ring, so a Clifford algebra—which needs a commutative scalar ring—can only be built over the field-like subclasses of game-world: the surreal numbers **No**, the surcomplex numbers **No**[i], and the nimbers On_2 . We describe **pleroma**, a small library realising Clifford algebras (with nilpotents) over all three, and focus on the characteristic-2 nimber case, where the Clifford algebra is classified by the *Arf invariant* of an $\mathbb{F}_2$ quadratic form (Ovsienko). This is an experimental note. Its modest contributions are: (i) an implementation that keeps the quadratic and polar forms independent, so the char-2 theory is faithful, and that computes the Arf invariant over any nim-subfield via the field trace; (ii) the observation, verified computationally, that the Arf-bearing Gold forms $\text{Tr}(x^{1+2^a})$ are *composites of combinatorial-game operations*; and (iii) the framing of the Arf invariant as a *win-bias*: by Dickson’s democratic-value theorem it is exactly the sign of the majority margin of the form’s zero-set, which we confirm on the game-built forms. We close with a sharp open question—is there a *natural* game whose P -positions are such a form’s zero set?—and narrow it with two computational probes: an interactive-game kernel solver and a (bounded) misère indistinguishability-quotient computer. These pin the obstruction to the polar form, separate a tautological “solution” (a rule that references Q itself) from a genuine one, and, via a sweep of 292 octal games, rule out the one misère-quotient shape that could host such a zero set as a quadric in the range searched. We then prove a structural no-go that explains these negatives uniformly: a game whose move rule sees only the symplectic form B is $\text{Sp}(B)$ -equivariant, hence—as $\text{Sp}(B)$ is transitive on nonzero vectors—can have no quadric as its P -set; a game that also reads the coordinate frame does realise a genuine quadric, but the *split* one, and the gap to the Gold form is exactly a *framing*, the diagonal $q_i = \text{Tr}(e_i^{1+2^a})$, which we find always carries $\text{Arf} = 1$. The classical results we lean on (the Arf–Clifford correspondence, Dickson’s theorem, the Gold-function ranks, and Turning-Corners = nim-multiplication) are reproduced by the tool as validation, not claimed as new.

1 Why three backends

A (short, partizan) combinatorial game is a pair $\{L \mid R\}$ of sets of previously-constructed games; under the disjunctive sum the games form a partially ordered abelian group [4, 3]. They do *not* form a ring: Conway’s game product is only a congruence on the *numbers*, the surreal subclass **No**. A Clifford algebra is a quotient of a tensor algebra and requires a commutative ring of scalars, so “Clifford over games” is forced onto the field-like cores of game-world:

core	structure	Clifford flavour
No (surreals)	real-closed field, char 0	the $\text{Cl}(p, q)$ theory over $\mathbb{R}$, exotic scalars
No [i] (surcomplex)	algebraically closed, char 0	the complex Clifford theory
On_2 (nimbers)	algebraically closed, char 2	genuinely different

No is real-closed [4], so finite-dimensional Clifford algebras over it are classified exactly as over $\mathbb{R}$ (the 8-fold periodicity), but one may now use infinite or infinitesimal metric entries such as $e_0^2 = \omega$, $e_1^2 = \varepsilon = \omega^{-1}$. **No**[i] is algebraically closed and gives the complex theory. The numbers On_2 —the ordinals under nim-addition and nim-multiplication—form an algebraically closed field of *characteristic* 2 [4], with the finite subfields $\mathbb{F}_{2^{2^k}}$ consisting of the numbers below 2^{2^k} . This is the only core where Clifford theory acquires a new character, and it is our focus.

2 Characteristic 2, faithfully

In characteristic 2 a quadratic form is not determined by a symmetric bilinear form. Writing $q_i = Q(e_i)$ for the squares and $b_{ij} = Q(e_i + e_j) + Q(e_i) + Q(e_j)$ for the polar form, the relations defining the algebra are

$$e_i^2 = q_i, \quad e_i e_j + e_j e_i = b_{ij},$$

and the polar form is *alternating* ($b_{ii} = 0$) while q_i may be nonzero. If one collapses (q, b) into a single symmetric bilinear form, every char-2 algebra over an orthogonal basis becomes commutative and the content is lost. **pleroma** therefore stores q and b as independent data and reduces products by a word-rewriting routine whose only use of the sign is through the scalar's own negation; since $-1 = 1$ in On_2 , the char-2 sign-vanishing falls out automatically.

For a nonsingular quadratic form Q on $\mathbb{F}_2^{2^m}$ with symplectic basis $\{a_k, b_k\}$, the *Arf invariant* is $\text{Arf}(Q) = \sum_k Q(a_k) Q(b_k) \in \mathbb{F}_2 [1]$; it is independent of the basis and is a complete invariant. The crucial fact for us is the classification it controls.

Theorem (Ovsienko [9]). The classification of (real) Clifford algebras is equivalent to the classification of quadratic forms over $\mathbb{F}_2$, with the Arf invariant the complete invariant.

So computing Arf of a nim-Clifford form returns the isomorphism class of the algebra: it is the char-2 Clifford classifier, not a toy. **pleroma** computes it for a form over any nim-subfield $\mathbb{F}_{2^{2^k}}$ by symplectic reduction over the field, then pushes the result to $\mathbb{F}_2$ via the trace $\text{Tr}_{\mathbb{F}_{2^m}/\mathbb{F}_2}(x) = x + x^2 + \dots + x^{2^{m-1}}$, which realises the canonical isomorphism $k/\wp(k) \cong \mathbb{F}_2$. Over $\mathbb{F}_4$, for instance, the form with $q = (*2, *2)$ is of type O^- and $q = (*2, *3)$ is hyperbolic.

3 The bridge to games

Nim-addition is the disjunctive sum (Sprague–Grundy), and nim-multiplication is itself a game. Conway's *Turning Corners* has, for the single coin at (x, y) , the Grundy value given by the excludant recurrence

$$x \otimes y = \text{mex}\{ (i \otimes y) \oplus (x \otimes j) \oplus (i \otimes j) : 0 \leq i < x, 0 \leq j < y \}, \quad (1)$$

which is Conway's definition of nim-multiplication [4, 3]. We implement (1) directly and verify it agrees with the algebraic Fermat-power product (computed via $F_n \otimes F_n = \frac{3}{2}F_n$ for Fermat 2-powers $F_n = 2^{2^n}$) on all pairs $x, y < 48$: two independent definitions, one combinatorial and one field-theoretic, of the same product.

This makes three operations game-realizable: the nim-product $\otimes$ (Turning Corners), the Frobenius $\square: v \mapsto v^2 = v \otimes v$ (the diagonal of Turning Corners), and $\oplus$ (disjunctive sum); hence also the trace, which is iterated $\square$ and $\oplus$.

Gold forms, and a validation. The natural quadratic forms on a char-2 field are $Q_a(x) = \text{Tr}(x \cdot x^{2^a}) = \text{Tr}(x^{1+2^a})$, the *Gold forms* [6]: $g = \text{Frob}^a$ is additive, so Q_a is genuinely quadratic. As a standard cross-check, the polar-form rank our classifier computes reproduces the known Gold-function rank $\text{rank} = m - 2 \gcd(a, m)$ (in the regime $m = 2^k$, where $m/\gcd$ is even) [8]; see Table 1.

field	a	Q	Arf	type	rank	$m - 2 \gcd(a, m)$
$\mathbb{F}_{2^4}$	1	$\text{Tr}(x^3)$	1	O^-	2	2
$\mathbb{F}_{2^8}$	1	$\text{Tr}(x^3)$	1	O^-	6	6
$\mathbb{F}_{2^8}$	2	$\text{Tr}(x^5)$	1	O^-	4	4
$\mathbb{F}_{2^{16}}$	1	$\text{Tr}(x^3)$	1	O^-	14	14
$\mathbb{F}_{2^{16}}$	4	$\text{Tr}(x^{17})$	1	O^-	8	8
$\mathbb{F}_{2^{32}}$	1	$\text{Tr}(x^3)$	1	O^-	30	30

Table 1: The classifier’s rank matches the Gold-function formula (15/15 cases up to $m = 32$; a representative selection).

The Gold forms are game-built. Because $\otimes$, $\square$, $\oplus$, and the trace are all game operations, the Gold form $Q_a(v) = \text{Tr}(v \otimes v^{2^a})$ is a composite of combinatorial-game operations applied to a position’s nimber value v (and under the 1-dimensional coin-turning game with single-coin Grundy value $g(n) = 2^n$, a position’s value *is* a nimber). We confirm this by rebuilding Q_a from literal Turning-Corners products and checking equality with the field computation on the small fields where the recurrence (1) is tractable.

Observation 1. *The quadratic forms whose Arf invariants classify the nim-Clifford algebras are not merely defined over the value field of games; they are composites of game operations. The char-2 Clifford theory is, in this constructive sense, made of games.*

4 The Arf invariant as a win-bias

The semantic payoff comes from a classical counting fact.

Theorem (Dickson [5]; Arf [1]). Over $\mathbb{F}_2$, the Arf invariant of a quadratic form is the value the form takes most often. For a nonsingular form on $\mathbb{F}_2^{2m'}$,

$$\#\{v : Q(v) = 0\} = 2^{2m'-1} + (-1)^{\text{Arf}(Q)} 2^{m'-1}.$$

Read game-theoretically: if a game had P -positions (second-player wins) exactly $\{v : Q(v) = 0\}$, then $\text{Arf}(Q)$ would be precisely the *sign of the win-bias*—which player wins from more starting positions—with a fixed margin of $2^{m'-1}$, a Gauss-sum, i.e. a square-root-scale fluctuation about 50/50. The Arf invariant is the second-order tiebreaker of an otherwise near-fair game.

We verify the count directly on the game-built Gold forms, handling the radical: for rank r and an isotropic radical, $\#\{Q = 0\} = 2^{m-1} + (-1)^{\text{Arf}} 2^{m-r/2-1}$ (and exactly 2^{m-1} when the radical is anisotropic). The brute-forced zero-counts match in every case (Table 2).

Observation 2. *On the game-built forms, the Arf invariant equals the win-bias in the counting sense: it is the sign of the majority margin of the zero-set.*

field	a	Arf	rank	$\#\{Q = 0\}$	bias
$\mathbb{F}_{2^4}$	1	1	2	4	-4
$\mathbb{F}_{2^8}$	1	1	6	112	-16
$\mathbb{F}_{2^8}$	2	1	4	96	-32
$\mathbb{F}_{2^{16}}$	1	1	14	32512	-256
$\mathbb{F}_{2^{16}}$	4	1	8	30720	-2048

Table 2: Zero-counts of the game-built Gold forms match $2^{m-1} + (-1)^{\text{Arf}} 2^{m-r/2-1}$ (bias = $\#\{Q = 0\} - 2^{m-1}$); 10/10 cases verified.

5 What remains open, and how far we narrowed it

The bridge closes at the level of *construction* (Observation 1) and of *counting* (the Arf invariant is the zero-set bias). What is not established is a *realisation*: a natural game whose P -positions are $\{v : Q(v) = 0\}$. The obstruction is structural and informative.

Remark 1. Normal-play disjunctive sums cannot do it. Their outcomes are $\oplus$ -linear in the components (Sprague–Grundy), so a P -position set is always a linear subspace, never a genuine quadric. This is exactly why coin-turning yields the *bilinear* nim-product but not a quadratic refinement.

The obstruction is exactly the polar form. In characteristic 2, $Q(u + v) = Q(u) + Q(v) + B(u, v)$, so for u, v already in the zero set ($Q(u) = Q(v) = 0$),

$$u + v \in \{Q = 0\} \iff B(u, v) = 0.$$

The failure of the candidate P -set to be a subspace is therefore measured, term for term, by the polar form B —and B is the *game-realizable* nim-product / coin-turning (Tartan) form. We confirm this concretely: the polar form of a Gold form, $B(e_i, e_j) = \text{Tr}(e_i \otimes e_j^{2^a} \oplus e_j \otimes e_i^{2^a})$, is reproduced exactly from literal Turning-Corners products on the bit basis. So of the three layers a P -position game would need—the *linear* part (Grundy/ $\oplus$), the *bilinear* coupling B , and a *quadratic* play rule—the first two are already built from games; the open piece is precisely a play rule that reads out Q rather than B .

A test bench. To interrogate any candidate, **pleroma** solves over $\mathbb{F}_2$ whether a set $S \subseteq \mathbb{F}_2^k$ is the zero set of *some* quadratic form (constant, linear, and quadratic terms, by Gaussian elimination), returning its Arf invariant and—crucially—whether it is a *genuine* quadric (nonzero polar rank) or a mere affine flat. As a sanity check it finds exactly 2^7 of the 2^8 subsets of $\mathbb{F}_2^3$ to be quadrics. Every P -set below is classified by this routine.

Two routes. Remark 1 leaves two escapes. *Misère* play first: misère sums are genuinely non-linear—the misère-Nim P -set is not $\{\oplus = 0\}$, not even a coset (Bouton [2]), clearing the bar normal play fails—and quadratic structure could live in a misère quotient [10, 3]. We implement a bounded indistinguishability-quotient computer (verified to return $\star \mapsto \mathbb{Z}/2$); the $\mathbb{F}_2$ -quadric question applies only when the quotient is an elementary abelian 2-group $(\mathbb{Z}/2)^k$. *Interactive* play second: a retrograde Win/Loss/Draw kernel solver for arbitrary finite game graphs lets us test any non-disjunctive move rule. Two findings sharpen the question rather than answer it. First, *existence is trivial*: a hand-built acyclic game has P -set exactly $\{Q = 0\}$, so the question was

never about existence but about a *natural* rule. Second, among uniform terminating rules on $\mathbb{F}_2^m$, the rule “move iff you flip Q ” does reproduce $\{Q = 0\}$ —but only by referencing Q in the move legality, i.e. tautologically; the rules coupled through the legitimate combinatorial datum B do *not* (a B -coupled descent gives an affine subspace, a single-bit B -gated turn a *different* quadric).

A hunt over octal games. The one quotient shape that could host a quadric P -set is $(\mathbb{Z}/2)^k$ with $k \geq 2$ and Arf-rank ≥ 2 . We swept 292 octal codes (length ≤ 3 , d_1 odd, heaps to 4; 876 bounded misère quotients in all), classifying each $(\mathbb{Z}/2)^k$ P -set with the test bench. The result is a clean negative (Table 3): the misère-quotient orders that occur are 2, 6, 10, 12, 14—*no power of 2 above 2 appears at all*, so no $(\mathbb{Z}/2)^{k \geq 2}$ arises, and there are no quadric P -sets.

misère-quotient order	# codes×cutoffs
2	432
6	433
10	6
12	2
14	3

Table 3: Bounded misère-quotient orders over 292 octal games (each at heap cutoffs $1..k$, $k = 2, 3, 4$). The only elementary-abelian-2 quotient is $\mathbb{Z}/2$ (order 2, a linear P -set); the wild quotients have orders 6, 10, 12, 14, never 4, 8, 16. No genuine quadric P -set occurs.

Observation 3. *Among the octal games searched, the only elementary abelian 2-group misère quotient is $\mathbb{Z}/2$, whose P -set is linear; the wild quotients are not 2-groups. The quadric P -set, if it exists, does not arise from an elementary-2-abelian octal misère quotient in this range.*

The symplectic obstruction. The interactive negatives above—a B -coupled descent collapsing to a subspace, a single-bit B -gated turn giving the wrong quadric—are not accidents but instances of one structural fact.

Proposition 1. *Let B be a nondegenerate alternating form on $V = \mathbb{F}_2^{2m'}$, and let a finite game have position set V and a move relation invariant under $\mathrm{Sp}(B)$ (in particular, any rule defined from B and the $\mathbb{F}_2$ -linear structure alone). Then its normal-play P -set is a union of $\mathrm{Sp}(B)$ -orbits on V , and since $\mathrm{Sp}(B)$ is transitive on $V \setminus \{0\}$ that set is one of $\emptyset, \{0\}, V \setminus \{0\}, V$ —never a genuine quadric.*

Proof. Each $g \in \mathrm{Sp}(B)$ is an automorphism of the move graph, and retrograde Win/Loss/Draw outcomes are determined by the graph alone, so the outcome function is $\mathrm{Sp}(B)$ -invariant and the P -set is a union of orbits. A symplectic group is transitive on the nonzero vectors of its space [11], so the orbits are $\{0\}$ and $V \setminus \{0\}$. A nondegenerate quadric has $2^{2m'-1} \pm 2^{m'-1}$ points, which is none of 0, 1, $|V| - 1, |V|$. $\square$

Over $\mathbb{F}_{2^4}$ we confirm the surrounding arithmetic directly: $|\mathrm{Sp}(4, 2)| = 720$, the orbit of any nonzero vector is all 15 nonzero vectors, and the set-stabiliser of a quadric $\{Q = 0\}$ is its orthogonal group $O(Q)$, of order 72 (type O^+ , orbit 10) or 120 (type O^- , orbit 6); the orbit sizes are exactly the numbers of Arf-0 and Arf-1 refinements of B ($10 + 6 = 16$).

B with a frame realises the split quadric. A game escapes $\text{Sp}(B)$ -equivariance as soon as it reads the standard basis. Define the *frame quadric* $Q_{\text{frame}}(v) = \sum_{i < j} B(e_i, e_j) v_i v_j$, built from B and the frame with zero diagonal. The all-downward rule “move $v \rightarrow w$ iff $w < v$ and $Q_{\text{frame}}(w) \neq Q_{\text{frame}}(v)$ ” has kernel exactly $\{Q_{\text{frame}} = 0\}$: any v with $Q_{\text{frame}}(v) = 1$ moves to 0, a terminal loss, so is a win; any v with $Q_{\text{frame}}(v) = 0$ reaches only wins, so is a loss. We verify this on 119/119 random nondegenerate B over $\mathbb{F}_{2^4}, \mathbb{F}_{2^6}, \mathbb{F}_{2^8}$. So a *natural* B -game does have a genuine quadric P -set—but Q_{frame} , whose Arf is fixed by (B, frame) .

The gap is a framing, and it carries the Arf. Two quadratic forms share a polar form iff they differ by a linear functional, so the refinements of B form a torsor under V^* . In particular $Q_a = Q_{\text{frame}} + \ell$, where $\ell(v) = \sum_i q_i v_i$ is the functional of the *diagonal* $q_i = Q_a(e_i) = \text{Tr}(e_i^{1+2^a})$, the single-coin self-Gold values. This is a *framing* in the Arf–Kervaire sense—the datum a symplectic form needs to become a quadratic one—and it is never free: for every genuinely-quadratic Gold form we tested (up to $\mathbb{F}_{2^{16}}$), $\text{Arf}(Q_{\text{frame}}) = 0$ while $\text{Arf}(Q_a) = 1$, so the diagonal always flips the split frame quadric to the Gold O^- . The whole $O^-/\text{Arf-1}$ content of the Gold form—the uniform O^- column of Table 1—is the framing; B and the frame contribute Arf 0. The obstruction is thus a ladder of symmetry-breaking, each rung a single datum:

$$\text{Sp}(B) \text{ (no quadric)} \xrightarrow{\text{frame}} O(Q_{\text{frame}}) \text{ (split quadric)} \xrightarrow{\text{diagonal}} O(Q_a) \text{ (Gold quadric)}.$$

Open question (sharpened). *The residue is a single named datum. By Proposition 1 no $\text{Sp}(B)$ -equivariant game realises any quadric; a game that reads the frame realises the split frame quadric (Arf 0); and the only thing between that and the Gold quadric is the m -bit diagonal framing $q_i = \text{Tr}(e_i^{1+2^a})$, not derivable from B . Is this framing itself game-natural—a coin marking, a misère or scoring twist, or a field-multiplicative coupling carrying the Frobenius non-locally—so that some natural game has P -set the Gold quadric $\{Q = 0\}$ with its Arf-1 win-bias? Referencing Q is tautological, the frame yields only Arf 0, and no $(\mathbb{Z}/2)^{k \geq 2}$ misère quotient appears among the octal games searched.*

A candidate must hit the win-bias of Table 2, and now also clear both the test bench and Proposition 1: its move rule must break the $\text{Sp}(B)$ symmetry down to $O(Q_a)$, i.e. encode the diagonal framing. The deepest relative, Welter’s game, lands in the representation theory of symmetric groups [7] rather than in Arf invariants.

Beyond the open thread. The library has also grown into a fuller Clifford engine: a characteristic-0 classifier (the companion to the Arf invariant, returning $\text{Cl}(p, q)$ as a matrix algebra over $\mathbb{R}, \mathbb{C}$, or $\mathbb{H}$ by 8-fold periodicity); general-bilinear-form (quantum-Clifford) products; the Witt group $W_q(\mathbb{F}_{2^m}) \cong \mathbb{Z}/2$; the Dickson invariant (the characteristic-2 determinant, with $\text{SO} = \ker D$); and the exterior algebra of the game *group* over $\mathbb{Z}$ —the one Clifford-adjacent structure that lives on *all* of game-world, non-numbers $(\star, \uparrow)$ included, since it needs only a module and not the (nonexistent) game product. These round out the engine; the open thread above remains its sharpest point.

The source, tests, and the experiments underlying all tables are available at <https://github.com/a9lim/pleroma>.

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